

Inference for Regression Slope

Given by excel sheet:

$$\hat{y} = 21.973 - 0.04249(\text{Minutes PA})$$

$$b = -0.04249$$

$$(-0.04583, -0.03914)$$

Hypotheses

$$H_0: \beta = 0$$

$$H_a: \beta < 0$$

Test Statistic

$$t = \frac{b - 0}{SE_b}$$

1. Margin of Error

$$CI: (-0.04583, -0.03914)$$

$$\text{Center: } \frac{-0.04583 + (-0.03914)}{2}$$

$$\approx -0.04249$$

$$ME = -0.03914 - (-0.04249) = 0.00335$$

2. t^* with 95% confidence $df = 59$

$$t^* = 2.001$$

$$ME = t^* \cdot SE_b \quad SE_b = \frac{ME}{t^*}$$

$$SE_b = \frac{0.00335}{2.001} \approx 0.00167$$

3. Test Statistic

$$t = \frac{b}{SE_b}$$

$$t = \frac{-0.04249}{0.00167} \approx -25.43$$

P-Value

$$t = -25.43 \quad df = 59$$

$$t_{cdf}(-9999, -25.43, 59)$$

$$\approx 1.1 \times 10^{-34}$$

$$p < 0.0001$$

95% Confidence Interval

$$b \pm t^* SE_b$$

$$-0.04249 \pm 2.001(0.00167)$$

Lower bound:

$$-0.04249 - 0.00334 = -0.04583$$

Upper bound:

$$-0.04249 + 0.00334 = -0.03915$$

$$(-0.04583, -0.03914)$$

One proportion z test

$$\hat{p} = \frac{7}{60} = 0.1167$$

$$p_0 = 0.246$$

$$\text{normcdf}(-99, -2.33, 0, 1) \approx$$

$$H_0: p = 0.246$$

$$H_a: p < 0.246$$

$$p = 0.010$$

$$0.010 < 0.05$$

↓
reject H_0

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{0.1167 - 0.246}{\sqrt{\frac{0.246(1-0.246)}{60}}} \approx -2.33$$

Confidence Interval

$$0.1167 \pm 1.96 \sqrt{\frac{0.1167(0.8833)}{60}}$$

$$0.1167 \pm 0.0813$$

$$(0.035, 0.198)$$